

Unit - I

Introduction to Computational Chemistry

What is Computational Chemistry?

Computational chemistry is a branch of chemistry that uses computer simulations and mathematical models to solve chemical problems.

- Normal chemistry is done in labs: mixing liquids, heating, using test tubes.
- But lab experiments can be slow, expensive, or even dangerous.
- Instead, in computational chemistry, scientists use computers to act like virtual labs.

Why Do We Need Computational Chemistry?

Think about designing a new car.

- In the past, engineers made physical models and tested them in real crash tests.
- Today, they first test cars using computer simulations (like in video games).
- This is faster, cheaper, and safer.

Similarly, in chemistry:

- Making a new chemical or drug in a lab takes years and crores of rupees.
- But a computer can predict results in hours or days.

Scope of Computational Chemistry

Computational chemistry is used in many areas:

1. Biochemistry (Inside our body)

- Helps us study proteins (the body's machines) and how they work.
- Used to see how enzymes (special proteins that help digest food) do their job.
- Helps us understand DNA (the code of life) and how drugs or chemicals affect it.
- Can predict what happens if there is a change in a gene.

Example: Studying how COVID-19 virus attaches to human cells.

Example: Designing better medicines for diseases like diabetes or cancer.

2. Materials Science:

- Helps design better batteries for mobiles and electric cars.
- Used to create stronger metals for planes and buildings.
- Helps improve solar panels to capture more sunlight.
- Used to make faster chips for computers and phones.

Example: Tesla uses simulations to design new batteries.

Example: Scientists test new plastics on computer before making them.

3. Environmental Chemistry:

- Helps us see how pollution spreads in air, water, or soil.
- Shows how harmful gases (like CO₂) affect climate and global warming.
- Helps design eco-friendly chemicals that are safe for nature.
- Predicts how plastics and pesticides break down in the environment.

Example: Checking how river water is affected by chemical waste.

Example: Understanding how ozone layer gets damaged.

4. Drug Discovery:

- Computers can test millions of drug molecules quickly.
- Shows which drug will fit best into the body's proteins (like a key in a lock).
- Helps make drugs more safe, soluble, and effective.
- Saves a lot of time and money in making new medicines.

Example: COVID-19 vaccines and antiviral drugs were first tested on computers.

Example: Cancer drugs are checked virtually before lab testing.

5. Agriculture and Food:

Helps design better fertilizers that give higher crop yield.

Predicts how pesticides will behave in soil and plants.

Used to make better food preservatives that are safe to eat.

Helps in making plant-based medicines and vitamins.

Example: Finding natural compounds in turmeric or neem that fight germs.

6. What is Molecular Geometry?

Bond Length (distance between two atoms)

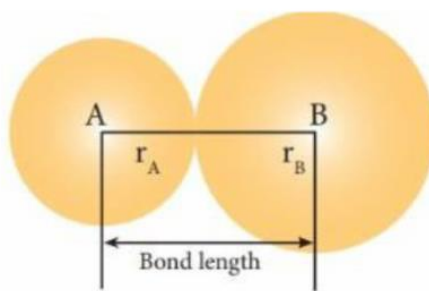
Bond length = the distance between the centers of two connected atoms.

the equilibrium distance between the nuclei of two bonded atoms in a molecule.

Shorter bond = stronger connection.

Longer bond = weaker connection.

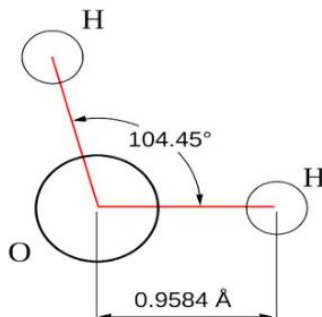
Example: In water, the O-H bond length is about 0.096 nanometers (very, very small!).



Bond length of covalent molecule A – B

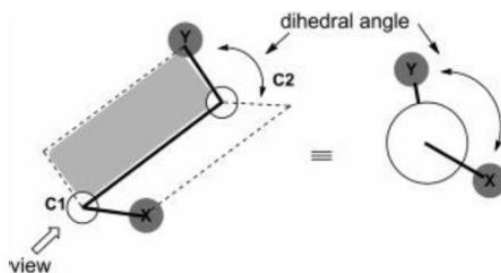
Bond Angle (angle between two bonds)

- When three atoms are connected, they form an angle.
- The angle between two covalent bonds originating from the same central atom in a molecule, measured in degrees.



Dihedral Angle (twist angle between two planes)

- A dihedral angle is the angle between two intersecting planes or half-planes
- Imagine four atoms connected one after another: A-B-C-D.
- The angle depends on how the front part of the molecule is twisted compared to the back part.



What is Geometry Optimization?

Geometry optimization is a process in which the shape of a molecule is adjusted step by step using a computer until it finds the most stable structure.

- Atoms in a molecule are not fixed; they can move like balls joined by springs.
- The computer keeps moving the atoms slightly, again and again, until they settle in a shape where the molecule feels most relaxed (lowest energy).
- This is like the molecule finding its most stable position.
- Molecules prefer to exist in their lowest-energy form because it is most stable.
- Geometry optimization finds this stable structure by reducing the "stress" in the bonds. Example: Water (H₂O) naturally bends into a Mickey Mouse shape, not a straight line.
- A transition state is a temporary, high-energy shape the molecule passes through when moving from one stable form to another.
- Geometry optimization can also help find these transition states.
- By knowing both the stable shapes and the transition states, scientists can understand how and why reactions happen.

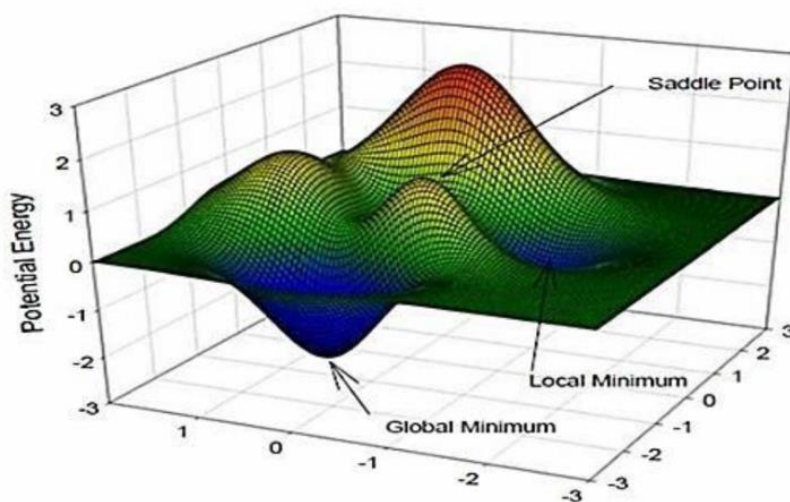
- This helps explain why some reactions are fast, some are slow, and why certain
- molecules are more reactive than others.

Potential Energy Diagram

Explanation of the Graph (Potential Energy Surface)

- This 3D graph shows how the energy of a molecule changes with different atomic arrangements (geometries).
- X and Y axes (horizontal): Different possible shapes/positions of the molecule.
- Z axis (vertical): Potential energy of the molecule at that shape.

The surface looks like hills and valleys.



1. Global Minimum

- This is the deepest valley (lowest point on the graph).
- It represents the most stable structure of the molecule.
- The molecule naturally prefers to sit here.

2. Local Minimum

- These are small valleys (not as deep as the global minimum).
- They represent other possible stable shapes of the molecule, but not the best one.
- A molecule can get "trapped" here if it doesn't have enough energy to move to the global minimum.

3. Saddle Point (Transition State)

- This is like the top of a mountain pass between two valleys.
- It represents a temporary, unstable structure that molecules pass through during a chemical reaction.
- It's not stable → the molecule won't stay here long.

The lowest point (valley) on the PES represents the most stable structure or global minimum energy conformation. Saddle points on the PES correspond to transition states - high-energy intermediates in chemical reaction. PES helps in understanding reaction mechanisms, transition states, and stable molecular conformations. Computational chemistry methods are used to calculate energies for different molecular geometries and construct the PES.

Methods in Computational Chemistry

Methods in computational chemistry refer to the mathematical models and algorithms used to simulate, calculate, and predict the structure, properties, reactions, and behavior of molecules and materials using computers.

Methods can be broadly classified as:

Quantum Mechanical Methods: Solve the Schrödinger equation to describe electrons and atomic interactions.

Classical Methods: Use Newtonian mechanics and force fields to model molecular structures. Examples: AMBER, CHARMM, OPLS force fields.

Hybrid Methods: Combine quantum mechanics for reactive sites with classical mechanics for the rest of the system.

Some modern methods used in computational chemistry are:

1. Ab Initio Methods

Based purely on quantum mechanics without using any empirical data.

"Ab Initio" is Latin for "from the beginning."

These methods use pure quantum mechanics without relying on experimental data.

They solve the Schrödinger equation (the main equation of quantum mechanics) to describe how electrons behave in molecules.

Features: Very accurate because they calculate everything from scratch. Very slow and expensive in terms of computer power. Usually used only for small molecules because larger systems take too much time.

Pros: Most accurate results. Can predict new properties that are not yet measured in experiments.

Cons: Very slow and needs powerful computers. Not practical for very large systems (like proteins).

Applications: Understanding small molecules deeply. Predicting bond lengths, energies, and reactions with high accuracy.

2. Density Functional Theory (DFT)

- Instead of focusing on the full wavefunction of all electrons (which is very complex), DFT uses electron density (a simpler property).
- This makes the calculations faster and easier, while still giving good accuracy.

Features: A balance between accuracy and speed. Less expensive than Ab Initio, but more reliable than Semi-Empirical methods. Widely used in chemistry, physics, and materials science

Pros: Accurate for most molecules. Can handle bigger systems (compared to Ab Initio). Popular and widely used.

Cons: Not as perfect as Ab Initio. Sometimes less reliable for very complex reactions

Applications:

Studying molecular geometry. Reaction energies. Designing new materials and catalysts.

3.Semi-Empirical Methods

- These methods are also based on quantum mechanics but use experimental data to simplify calculations.
- Instead of calculating everything, they take shortcuts by fitting some parts to known experimental values.

Features: Much faster than Ab Initio or DFT. Less accurate because of the approximations. Good for large molecules where full quantum treatment is impossible.

Pros: Very fast. Can handle very large molecules

Cons:

- Accuracy depends on the quality of experimental data used. Not suitable for very precise predictions.

Applications:

- Screening large numbers of molecules. Initial studies before using more accurate methods.

4. Molecular Mechanics (MM)

- Uses classical physics (Newton's laws) instead of quantum mechanics.
- Molecular Mechanics is a classical approach used to model the structure, energy, and dynamics of molecules by treating atoms as balls and bonds as springs. Instead of solving quantum equations, it applies Newton's laws of motion to simulate molecular systems.
- Molecular mechanics focuses on bond lengths, bond angles, torsional angles and non-bonded interactions (like van der Waals and electrostatic forces).
- The total potential energy of a molecule is calculated using a force field formula.

$$E_{total} = E_{bonds} + E_{angles} + E_{torsions} + E_{non-bonded}$$

Where, E_{bonds} , E_{angles} , $E_{torsions}$ and $E_{non-bonded}$ represent the bond stretching energy, angle bending energy, torsional (dihedral) angle energy and the non-bonded interaction energy, respectively.

- Force fields are predefined sets of parameters and equations used to describe molecules.
- Atoms are treated as balls, and bonds are like springs connecting them.
- Uses force fields (mathematical models) to describe interactions like bond stretching, bending, or repulsion.

Features:

- Very fast and efficient. Cannot describe electrons, so it cannot handle reactions (bond breaking/forming). Works best for very large biological molecules like proteins, DNA, or polymers.

Pros:

- Extremely fast. Can handle very large systems (proteins, DNA, materials). Useful for long simulations (molecular dynamics).

Cons:

- Ignores electrons → cannot study reaction chemistry. Less accurate for transition states and charged species.

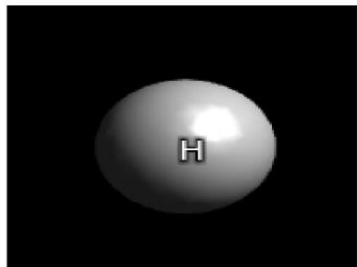
Applications:

- Protein structure modeling. Drug design (docking studies). Material property predictions.

➤ Representing a molecule using Z-matrix

- A Z-matrix provides a description of each atom in a molecule in terms of its atomic number, bond length, bond angle and the dihedral angle.
- Z-matrix representation is also called the internal coordinate representation.

Hydrogen atom

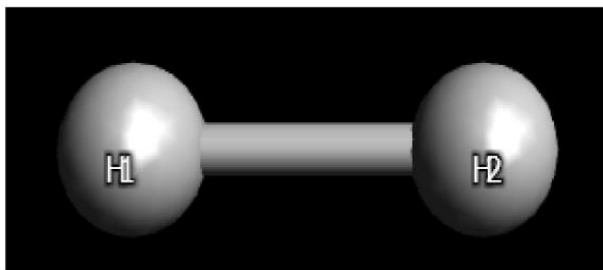


Z-matrix:

H

The hydrogen atom can be represented by just writing the symbol for hydrogen, 'H', as there are no bond lengths or bond angles involved (no internal coordinates).

Hydrogen Molecule



Z-matrix:

H

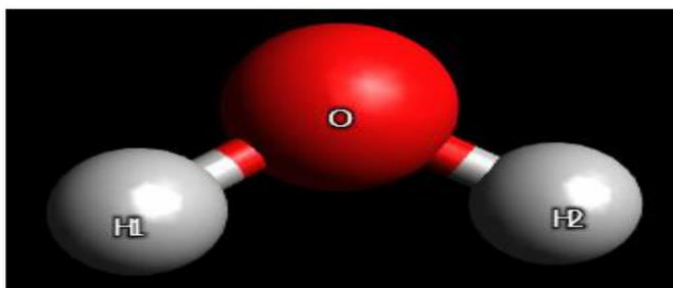
H 1 r2

Variables:

r2= 1.4352

- The first hydrogen atom is represented by the symbol 'H' in the first line.
- The representation of the second hydrogen atom is given in the second line of the Z-matrix. This should be interpreted as, the atom given in the second line (H) is at a distance of r2 (r2= 1.4352 Å) from the atom given in the first line (the first hydrogen atom).

Water molecule



Z-matrix:

O

H 1 r2

H 1 r3 2 a3

Variables:

r2= 0.9700

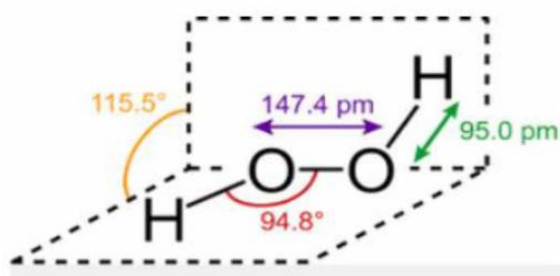
r3= 0.9700

a3= 109.47

Ref notes: Archana S

- The oxygen atom is represented by the symbol 'O' in the first line.
- The representation of the first hydrogen atom is given in the second line of the Z-matrix. This should be interpreted as, the atom given in the second line (H) is at a distance of r_2 ($r_2 = 0.9700 \text{ \AA}$) from the atom given in the first line (the oxygen atom).
- The second hydrogen atom is represented in the third line. This should be interpreted as, the atom in the given line (the second hydrogen atom) is at a distance of r_3 ($r_3 = 0.9700 \text{ \AA}$) from the atom in the first line (O) and makes an angle a_3 ($a_3 = 109.47^\circ$) with the atom in the first line (O) and the one in the second line (H).

Hydrogen peroxide (H₂O₂) molecule



Z-matrix:

```
O
O 1 r2
H 1 r3 2 a3
H 2 r4 1 a4 3 d4
Variables:
r2= 1.4740
r3= 0.9500
a3= 94.8
r4= 0.9500
a4= 94.8
d4= 115.5
```

- The first oxygen atom is represented by the symbol 'O' in the first line.
- The representation of the second oxygen atom is given in the second line of the Z-matrix. This should be interpreted as, the atom given in the second line (O) is at a distance of r_2 ($r_2 = 1.4740 \text{ \AA}$) from the atom given in the first line (the first oxygen atom).
- The first hydrogen atom is represented in the third line. This should be interpreted as, the atom in the given line (the first hydrogen atom) is at a distance of r_3 ($r_3 = 0.9500 \text{ \AA}$) from the atom in the first line (O) and makes an angle a_3 ($a_3 = 94.8^\circ$) with the atom in the first line (O) and the one in the second line (H).
- The H in the fourth line represents the second hydrogen atom. This line tells us that the second hydrogen atom is at a distance of r_4 ($r_4 = 0.9500 \text{ \AA}$) from the atom in the second line (the second oxygen atom), make an angle of a_4 ($a_4 = 94.8^\circ$) with the atoms in the second and the first lines and finally makes a dihedral angle, d_4 ($d_4 = 115.5^\circ$) with the atoms in the first line (O), the second line (O) and the third line (H).